Practical – 7

Aim:

To model a simple Bayesian Network or probabilistic classifier using a suitable Python library and perform probabilistic inference for prediction or diagnosis based on given data.

Code:

```
from sklearn.naive_bayes import BernoulliNB

x=[[1,1,1],
   [1,1,0],
   [1,0,1],
   [0,1,0],
   [0,1,1],
   [0,0,0]
   ]
y=[
    'Flu',
    'Flu',
    'Flu',
    'Cold',
    'Cold',
    'Cold',
]

model = BernoulliNB()
model.fit(x,y)
patient =[[1,1,1]]
prediction =model.predict(patient)
print("Predicted diagnosis:",prediction[0])

prbabilities = model.predict_proba(patient)
for diagnosis, probability in zip(model.classes_, prbabilities[0]):
    print(diagnosis, "=", round(probability, 2))
```